Solving Systems by Substitution
Answer Key
Algebra 1

Quick Answers

1. $x = 3$, $y = 6$
2. $x = 1$, $y = 3$
3. $x = 3$, $y = 5$
4. $x = 5$, $y = 4$
5. $x = 4$, $y = 1$
6. $x = 3$, $y = 1$
7. (a) solution = $\emptyset$, —
8. $x = 7$, $y = 2$
9. (a) second equation solved for y = $3x - 4$, —
10. $x = 2$, $y = -1$
11. $x = 20$, $y = 25$
12. (a) solution = $x = 6$, $y = 5$, —

What is verified

9 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 7, 9, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSA-REI.C.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 15 tagged checks.

1. $y = x + 3,\ \ 2x + y = 12.$
Solution: The first equation already gives y, so put $x + 3$ into the second in place of y: $2x + (x + 3) = 12$, so $3x = 9$ and $x = 3$. Substituting back, $y = 3 + 3 = 6$. Check: $2(3) + 6 = 12$. $\boxed{(3,\ 6)}$

2. $x = y - 2,\ \ 3x + 2y = 9.$
Solution: Here it is x that is ready. Substituting into the second equation: $3(y - 2) + 2y = 9$, so $3y - 6 + 2y = 9$, $5y = 15$, $y = 3$. Then $x = 3 - 2 = 1$. Check: $3(1) + 2(3) = 9$. $\boxed{(1,\ 3)}$

3. $y = 2x - 1,\ \ 4x - y = 7.$
Solution: Substituting, and keeping the brackets so the minus sign reaches both terms: $4x - (2x - 1) = 7$, so $4x - 2x + 1 = 7$, $2x = 6$, $x = 3$. Then $y = 2(3) - 1 = 5$. Dropping those brackets and writing $4x - 2x - 1$ is the usual slip. $\boxed{(3,\ 5)}$

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4. $x + y = 9,\ \ 3x - y = 11.$

Solution: Neither equation is ready. The first has coefficients of 1, so isolate there: $y = 9 - x$. Substituting into the second: $3x - (9 - x) = 11$, so $3x - 9 + x = 11$, $4x = 20$, $x = 5$, and $y = 9 - 5 = 4$. Check: $3(5) - 4 = 11$. $\boxed{(5,\ 4)}$

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5. $y = -2x + 9,\ \ y = x - 3.$

Solution: Both sides equal y, so they equal each other: $-2x + 9 = x - 3$, giving $12 = 3x$ and $x = 4$. Then $y = 4 - 3 = 1$. Check in the first equation: $-2(4) + 9 = 1$. $\boxed{(4,\ 1)}$

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6. $2x + y = 7,\ \ 3x - 4y = 5.$

Solution: The y in the first equation has coefficient 1, so isolating there avoids fractions: $y = 7 - 2x$. Substituting into the second:

$$\begin{aligned} 3x - 4(7 - 2x) &= 5 \\ 3x - 28 + 8x &= 5 \\ 11x &= 33 \ \Rightarrow\ x = 3, \end{aligned}$$

and $y = 7 - 2(3) = 1$. Check: $3(3) - 4(1) = 5$. $\boxed{(3,\ 1)}$

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7. $y = 4x - 2,\ \ 8x - 2y = 10$: (a) solve; (b) what does the last line mean?

Solution (a): Substituting into the second equation: $8x - 2(4x - 2) = 10$, so $8x - 8x + 4 = 10$, which reduces to $4 = 10$. Every x has vanished and the statement left behind is false. $\boxed{\text{no solution}}$

Solution (b) — instructor-judged. Because $4 = 10$ is false, there is no pair (x, y) that satisfies both equations at once. On a graph, the second equation rearranges to $y = 4x - 5$: same slope of 4 as the first, different intercept, so the two lines are parallel and never cross. Full credit connects the false statement to there being no solution.

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8. $x - 3y = 1,\ \ 2x + 5y = 24.$

Solution: The x in the first equation has coefficient 1: $x = 3y + 1$. Substituting: $2(3y + 1) + 5y = 24$, so $6y + 2 + 5y = 24$, $11y = 22$, $y = 2$, and $x = 3(2) + 1 = 7$. Check: $7 - 3(2) = 1$ and $2(7) + 5(2) = 24$. $\boxed{(7,\ 2)}$

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9. $y = 3x - 4,\ \ 6x - 2y = 8$: (a) solve the second for y; (b) how many solutions?

Solution (a): Subtract $6x$ from both sides and divide by -2, or rearrange as $2y = 6x - 8$ and halve: $y = 3x - 4$. $\boxed{y = 3x - 4}$

Solution (b) — instructor-judged. Substituting the first equation into the second gives $6x - 2(3x - 4) = 8$, that is $6x - 6x + 8 = 8$, or $8 = 8$ — true for every value of x. Part

(a) explains why: the second equation *is* the first, written differently, so the two graphs are one line and every point on it satisfies both. The system has infinitely many solutions. Full credit ties the always-true statement to the two equations being the same line.

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10. $3x + 2y = 4, \;\; x - y = 3.$

Solution: The second equation isolates cleanly: $x = y+3$. Substituting: $3(y+3)+2y = 4$, so $3y + 9 + 2y = 4$, $5y = -5$, $y = -1$, and $x = -1 + 3 = 2$. Check: $3(2) + 2(-1) = 4$.

$\boxed{(2, -1)}$

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11. Booth: 45 tickets, 285 dollars, adults 8 dollars, children 5 dollars.

Solution: Counting tickets gives $x + y = 45$; counting money gives $8x + 5y = 285$. Isolate from the count equation, where both coefficients are 1: $y = 45 - x$. Substituting into the money equation:

$$\begin{aligned} 8x + 5(45 - x) &= 285 \\ 8x + 225 - 5x &= 285 \\ 3x &= 60 \;\Rightarrow\; x = 20, \end{aligned}$$

so $y = 45 - 20 = 25$. Twenty adult tickets and twenty-five child tickets. Check: $20 + 25 = 45$ and $8(20) + 5(25) = 160 + 125 = 285$.

$\boxed{(20, 25)}$

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12. $y = \dfrac{2}{3}x + 1, \;\; x + y = 11$: (a) solve; (b) why substitution here?

Solution (a): The first equation is already solved for y, so substitute it into the second:

$$\begin{aligned} x + \tfrac{2}{3}x + 1 &= 11 \\ \tfrac{5}{3}x &= 10 \;\Rightarrow\; x = 6, \end{aligned}$$

and then $y = \frac{2}{3}(6) + 1 = 5$. Check: $6 + 5 = 11$.

$\boxed{(6, 5)}$

Solution (b) — instructor-judged. One equation is already in the form "y = an expression in x", which is exactly the shape substitution wants — one replacement and the system becomes a single equation in x. Elimination would first need the fraction cleared and the coefficients matched, which is more work for the same answer. Full credit points at one equation already being solved for a variable as the signal to substitute.